

24. Vinegar is the solution of which of the following?

- (a) Oxalic acid (b) Citric acid
(c) Hydrochloric acid (d) Acetic acid

U.P.P.C.S. (Pre) 2009

42nd B.P.S.C. (Pre) 1997

I.A.S. (Pre) 1993

Ans. (d)

Acetic acid (CH_3COOH) or ethanoic acid is an organic acid, which is responsible for sour taste and the pungent smell of the vinegar. The physical and chemical properties prove that the vinegar is a solution of acetic acid. It is produced from fermentation of ethanol.

25. Sour taste of vinegar is due to :

- (a) Tartaric acid (b) Lactic acid
(c) Citric acid (d) None of these

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (d)

See the explanation of above question.

26. Which acid is found in Vinegar?

- (a) Lactic Acid (b) Citric Acid
(c) Malic Acid (d) Acetic Acid

M.P.P.C.S. (Pre) 2010

Ans. (d)

Acetic acid is the chief component of vinegar. Vinegar is a liquid consisting mainly of acetic acid (CH_3COOH) and water. The acetic acid is produced by the fermentation of ethanol (formed from sugarcane juice and molasses) by acetic acid bacteria. It is mainly used as cooking ingredient and as a preservative.

27. Vinegar is chemically known as :

- (a) Acetic acid (b) Butyric acid
(c) Formic acid (d) Tartaric acid

M.P. P.C.S. (Pre) 2016

Ans. (a)

See the explanation of above question.

28. The chemical name for vinegar is :

- (a) Methanol (b) Ethanol
(c) Acetic acid (d) Ethyl acetate
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (c)

See the explanation of above question.

29. Which of the following statements is correct?

- A. Sugarcane juice is fermented to form vinegar.
B. Vinegar has acetic acid.

C. Benzoic acid is the acid present in Vinegar.

D. Vinegar is used as a preservative.

Code :

- (a) A, B, D (b) B, C, D
(c) A, C, D (d) A, B, C

R.A.S./R.T.S. (Pre) 2016

Ans. (a)

See the explanation of above question.

30. Which of the following is not optically active?

- (a) Glycine (b) Alanine
(c) Serine (d) All of the above

Jharkhand P.C.S. (Pre) 2013

Ans. (a)

Glycine is a non-essential, non-optical amino acid. It is also known as a building block for protein. Its formula is $\text{NH}_2\text{CH}_2\text{COOH}$. An optically active compound should have at least one carbon atom attached to four different groups. Glycine is not optically active because the carbon atom is attached to 2 hydrogen atoms, 1 amino group and 1 carboxylic group.

E. The Explosive Substances

Notes

- An explosive is a substance that can be made to explode by being hit or lit on fire.
- Explosives contain highly reactive chemicals that are full of potential energy, which can be released with a bang.
- A good explosive should contains following properties—
 - Explosive should not be volatile.
 - Explosive should not be hygroscopic.
 - Explosive should be stable.
 - Upon ignition, it decomposes rapidly in a detonation.
- Some explosives are as follows—

Trinitrotoluene (TNT) :

- TNT is made by adding nitric acid to toluene.
- It is a chemical compound with the formula $\text{C}_6\text{H}_2(\text{NO}_2)_3\text{CH}_3$.
- This is a yellow solid material.
- TNT is used in military shells, bombs and grenades. Its industrial uses include deep well and underwater blasting, chemical manufacturing as an intermediate in the production of dye-stuffs and photographic chemicals.
- Amatol** is an explosive made by mixing TNT with ammonium nitrate.

Trinitroglycerine (TNG) :

- It is a dense, colourless, oily, explosive liquid most commonly produced by nitrating glycerol with white fuming nitric acid in presence of sulphuric acid.
- It is also known as nitroglycerine or **Nobel oil**.
- Trinitroglycerine is a high explosive itself but is extremely shock-sensitive and degrades overtime to even more unstable forms.
- Trinitroglycerine is used to make **dynamite**.

Trinitrophenol (TNP) :

- It is also known as **picric acid** with the formula $(\text{O}_2\text{N})_3\text{C}_6\text{H}_2\text{OH}$.
- It is one of the most acidic phenols.
- It is made by the reaction of phenol and conc-nitric acid.
- It is mainly used as an explosive.
- It has also been used in medicine (antiseptic, burn treatments) and dyes.
- It is a light yellow, crystalline solid.

Dynamite :

- It was invented by Alfred Nobel in 1867.
- It is made by absorbing nitroglycerine in sawdust or kieselguhr (diatomaceous earth also known as D.E. or a naturally occurring soft, siliceous sedimentary rock that is easily crumbled into a fine white powder).
- Dynamite is mainly used in the mining, quarrying (excavation), construction and demolition industries.

Blasting Gelatin :

- Blasting gelatin is a mixture of 7% nitrocellulose and 93% of nitroglycerine.
- It is used in quarries and mining as an explosive.

RDX :

- RDX was discovered by George Friedrich Henning of Germany. It is an organic compound with the formula $(\text{O}_2\text{NNCH}_2)_3$.
- Its chemical name is cyclonite or cyclotrimethylene-trinitramine.
- It is also known as **plastic explosive**.
- RDX is a white solid without smell or taste, widely used as explosive.
- RDX is known as hexogen in Germany, T-4 in Italy and cyclonite in United States of America.
- It is the explosive agent in C-4 plastic explosive.

Gun cotton :

- Nitrocellulose or gun cotton is a highly flammable compound formed by nitrating cellulose (cotton or wood fibres) through exposure to nitric acid.

- It is a mild explosive, used in rockets, propellants, printing ink bases, leather finishing and celluloid. A mixture of nitrocellulose and campher, first used to manufacture **billiard balls**.

Question Bank

1. A mixture of which one of the following pairs of gases is the cause of occurrence of most of the explosions in mines?

- (a) Hydrogen and Oxygen
- (b) Oxygen and Acetylene
- (c) Methane and Air
- (d) Carbon dioxide and Methane

I.A.S. (Pre) 2008

Uttarakhand P.C.S. (Pre) 2007

Ans. (c)

Generally carbon monoxide, carbon dioxide, methane, oxygen, nitrogen and hydrogen gases etc. are found in mines. When combined with air, they cause an explosion. About 5–15% of the amount of methane in air is explosive. In air, about 12.5–74% presence of carbon monoxide is explosive. In air, about 4–74% carbon and hydrogen mixture may blast. Among the given options, in mines, mostly methane and air mixture causes explosions.

2. Which one of the following is not an explosive :

- (a) Trinitrotoluene (TNT)
- (b) Trinitroglycerine
- (c) Cyclotrimethylene Trinitramine (RDX)
- (d) Nitrochloroform

U.P.P.C.S. (Pre) 2014

U.P.P.C.S. (Pre) 2003

U.P.U.D.A./L.D.A. (Pre) 2002

U.P.P.C.S. (Pre) 2001

Ans. (d)

Nitrochloroform, also known as chloropicrin, is a chemical compound currently used as a broad-spectrum antimicrobial, fungicide, herbicide, and insecticide. It is also used as tear gas. It was manufactured for use as poison gas in World War I. Other options are explosives.

3. Which one of the following is not an explosive?

- (a) Nitrochloroform
- (b) TNG
- (c) TNT
- (d) Potassium Chlorate

U.P.P.C.S. (Mains) 2015

Ans. (a)

See the explanation of above question.

4. 'Nobel oil' is the name of which one of the following explosives?

- (a) TNG (b) TNP
(c) TNA (d) TNT

U.P. Lower Sub. (Pre) 2015

Ans. (a)

Trinitroglycerin (TNG) also known as Nobel oil, is an explosive fluid. Chemically, it is an organic nitrate compound.

5. RDX was invented by

- (a) Alfred Nobel (b) Soddy
(c) Bergillins (d) Henning

U.P. PCS (Mains) 2016

Ans. (d)

RDX, the abbreviation of Research Department Explosive, is also called cyclonite. This powerful explosive was discovered by George Friedrich Henning of Germany and patented in 1898 but not used until World War II, when most of the warring powers introduced it.

6. Which one of the following statement is correct?

- (a) DDT is antibacterial compound.
(b) TNT is an insecticide.
(c) RDX is an explosive.
(d) LSD is an antiviral compound.

U.P.P.C.S. (Pre) 1996

Ans. (c)

RDX (Research Department eXplosive or Royal Demolition eXplosive) is an explosive solid and its chemical formula is $(\text{CH}_2\text{N}_2\text{O}_2)_3$. Actually, RDX is cyclotrimethylenetrinitramine. It is also known as cyclonite, hexogen and T4. It is more powerful explosive than TNT. Due to its high explosive property, it was widely used in Second World War. RDX is a colourless solid substance. It is obtained by fuming nitric acid with hexamine.

7. Which one of the following is another name of RDX ?

- (a) Cyanohydrin (b) Dextran
(c) Cyclohexane (d) Cyclonite

I.A.S. (Pre) 2007

U.P.P.C.S. (Mains) 2007

Ans. (d)

See the explanation of above question.

8. What is/are the full form(s) of HMX?

1. High Melting Explosive.
2. High-Density Monoatomic Xenon.
3. Hedge Monetizing Xeno-currency.
4. Her Majesty's Explosive.

Select the correct answer using the codes given below.

- (a) 1 and 2 (b) 2 and 3

(c) Only 1

(d) 1 and 4

69th B.P.S.C. (Pre) 2023

Ans. (c)

HMX is a man-made, colourless liquid chemical which stands for High Melting Explosive. HMX, also called octogen, is a powerful and relatively insensitive nitroamine high explosive, chemically related to RDX. The compound's name is the subject of much speculation, having been variously listed as High Melting Explosive, High-velocity Military Explosive, or High-Molecular-weight RDX.

9. Which one of the following is used as an explosive?

- (a) Phosphorus Trichloride (b) Mercuric Oxide
(c) Graphite (d) Nitroglycerine

I.A.S. (Pre) 2009

Ans. (d)

Nitroglycerine is a heavy, colourless and oily liquid explosive. It is also known as trinitroglycerol (TNG), glyceryl trinitrate (GTN) or 1,2,3- trinitroxypropane.

10. Which one of the following explosives does not contain nitroglycerine as an essential constituent ?

- (a) Cordite (b) Blasting Gelatine
(c) Dynamite (d) Amatol

U.P.P.C.S. (Mains) 2010

Ans. (d)

Cordite, blasting gelatin and dynamite contain nitroglycerin as an essential constituent, while a highly explosive element is Amatol. It is composed of ammonium nitrate and TNT (Trinitrotoluene) in varying ratios.

11. Which one of the following is used in the production of explosives?

- (a) Glycerol (b) Methanol
(c) Urea (d) Oxalic acid

U.P. R.O./A.R.O. (Pre) 2016

Ans. (a)

Glycerol is an organic compound. Reacting with conc. HNO_3 and conc. H_2SO_4 , it produces nitroglycerin. Nitroglycerin is a heavy, colourless, oily, explosive liquid. It is also called trinitroglycerin (TNG), glyceryl trinitrate (GTN) or Nobel's oil. It is mainly used in the explosives such as dynamite.

12. The chemical used in manufacturing of dynamite is

- (a) Glycerol (b) Glycerol triacetate
(c) Glycerol trinitrate (d) Glycerol triiodate

Jharkhand P.C.S. (Pre) 2016

Ans. (c)

See the explanation of above question.